

Water Potential Bench

Finding out what water actually responds to, and why osmosis can stop while the two solutions are still at different concentrations.

Name _____

Date _____

Period _____

Demonstration: andresdbp.github.io/resources/demos/water-potential-bench/

1 Before you open anything

Answer from what you already know. Commit to it in ink — the point is to have something on paper that the demonstration can later confirm or contradict.

PREDICTION

A U-tube is divided at the bottom by a membrane that water crosses and solute never does. The left arm is filled with 5.0 mM sucrose, the right arm with pure water. Both arms start with liquid columns 150 cm tall. The tube is left alone until **net water movement stops**.

At that moment, what is the sucrose concentration in each arm?

Write one sentence saying what reasoning gave you those two numbers.

2 Setting up the tube and reading the six quantities

Open the demonstration. It starts on the **U-tube** tab, and it starts running. The table under the controls reports six quantities for each arm; every number you are asked for below is one of its rows.

1. Confirm the solute is set to **Sucrose**, $i = 1$.
2. Set **Temperature T** to **295 K**. The hint below the slider now reports a value of RT . Write it here, with units:
 $RT =$ _____
3. Set **Left arm** to **5.0 mM** and **Right arm** to **0.0 mM**.
4. Press **Refill level**. This returns both columns to 150.0 cm and pauses the run, so you can read the starting state. Fill in the two *start* columns below.

5. Press **Jump to equilibrium** and fill in the two *equilibrium* columns.

Row in the table	Left, start	Right, start	Left, equilibrium	Right, equilibrium
C (mM)				
Osmolarity iC (mOsm)				
Column height (cm)				
ψ_p (bar)				
ψ_s (bar)				
ψ (bar)				

a. At the start, the large readout labelled $\psi_{\text{left}} - \psi_{\text{right}}$ carries a sentence beneath it. Copy that sentence, and say which arm the water is leaving.

b. Exactly one of the six rows finished with the same value on both sides. Which row, and what value?

c. Nobody moved the ψ_p slider, because there is no ψ_p slider. Using your equilibrium column heights, explain in one sentence where the left arm's ψ_p of 0.194 bar came from.

d. Subtract to find the difference in column height, Δh . Multiply it by $0.000981 \text{ bar cm}^{-1}$. Now find $\psi_{s,\text{left}} - \psi_{s,\text{right}}$ from your table. Compare the two results.

Every value on this page is quoted at the temperature you set. RT appears inside $\psi_s = -iCRT$, so changing T changes every solute potential in the demonstration. T must be in kelvin for that expression to mean anything.

3 What actually equalises

One control, varied systematically. Keep the right arm at 0.0 mM, the solute on sucrose and T at 295 K throughout.

1. Set **Left arm** to the value in the first column.
2. Press **Refill level**, then **Jump to equilibrium**.
3. Read the two *Column height* entries and the two *C (mM)* entries. Read Δh from the dashed marker drawn between the two liquid surfaces on the diagram.
4. Repeat down the table.

Left arm set to	h left (cm)	h right (cm)	Δh (cm)	C left (mM)	C right (mM)	Ψ (bar)
2.0 mM						
4.0 mM						
6.0 mM						
8.0 mM						
10.0 mM						

a. Across all five runs, did the left and right concentrations ever come out equal? Did the gap between them get smaller or larger as you raised the left arm?

b. Pick any row. Compute $\Delta h \times 0.000981 \text{ bar cm}^{-1}$, and compute $RT(C_{\text{left}} - C_{\text{right}})$ with C in mol L^{-1} . Show both, then do it for a second row.

c. State, in your own words, the condition that has to be met before net water movement stops. Your sentence must not contain the word *concentration*.

d. Turn back to your prediction in Part 1. Was it borne out? If not, name the specific assumption in your reasoning that the data contradicts.

4 Using it the way a question would

Three short investigations. Predict before you press, every time.

A. Working backwards through *i*.

1. Select **NaCl**, $i = 2$. Set **Left arm** to **5.0 mM**, **Right arm** to **0.0 mM**, T still 295 K.
2. Before pressing anything: which row of your Part 3 table do you expect this run to reproduce? Write your answer first. _____
3. Press **Refill level**, then **Jump to equilibrium**.

Row read	Left	Right
C (mM)		
Osmolarity iC (mOsm)		
Column height (cm)		

a. Which sucrose run from Part 3 gave these same heights? Explain, in one sentence, what the two runs have in common that the molarity alone does not show.

B. Where the shortcut runs out. Go back to **Sucrose**, $i = 1$, left arm **5.0 mM**, right arm **0.0 mM**. Press **Refill level**, then **Jump to equilibrium**. The tube is now at rest.

b. You are about to press **Match the concentrations, keep the levels**. It stirs extra solute into the shorter arm until both arms read the same concentration, and it moves no water at all. Predict, before pressing: with the concentration difference removed, which way will water move?

1. Press the button and let the tube run until it settles by itself.
2. Record the settled state below.

Row read	Left	Right
Column height (cm)		
C (mM)		
Ψ_p (bar)		
Ψ (bar)		

c. At the instant you pressed the button the concentration gradient was exactly zero, and water moved anyway. Which term of $\Psi = \Psi_p + \Psi_s$ drove it, and in which direction?

d. The two arms have ended up at different concentrations again, even though nothing added or removed solute after the button was pressed. Explain how that happened.

C. One beaker, two answers. Switch to the **Cells** tab. Keep sucrose and 295 K. The animal cell holds 0.300 Osm; the plant cell holds 0.400 Osm at zero turgor.

1. Press **Fresh cells**, then the preset **The one that splits them**.
2. Let both cells settle, then fill in the table from the demonstration's own table and from the two tonicity readouts beneath it.

Row read	Bath	Animal	Plant
Osmolarity (Osm)			
Ψ_p (bar)			
Ψ_s (bar)			
Ψ (bar)			
Volume ($\times$ normal)	—		
Tonicity word shown	—		

e. One solution, two different tonicity words, at the same moment. What does that tell you about whether “hypertonic” can be a property of a liquid on its own?

f. Both cells reached the same final Ψ . The animal cell got there one way and the plant cell had a second route available. Name each route in terms of which component of Ψ the cell changed.

5 Solve these without the demonstration

Close the page, or look away from it. Show your working, and carry units. Then reopen it, check yourself, and if a number is off, write down which step produced the error. Use $R = 0.0831 \text{ L}\cdot\text{bar}\cdot\text{mol}^{-1}\cdot\text{K}^{-1}$ and $\rho g = 0.000981 \text{ bar per cm of water column}$.

1. Find Ψ_s for a 0.150 M NaCl solution in an open beaker at 300 K, then state Ψ for that solution. *To check: Cells tab, NaCl, Temperature $T = 300 \text{ K}$, Bath concentration = 0.150 M, read the Bath column.*

2. A plant cell is flaccid: $\Psi_p = 0$ and $\Psi_s = -9.81 \text{ bar}$. It is placed in an open beaker of 0.400 M sucrose at 295 K. Compute Ψ for the cell and Ψ for the solution, then predict the direction of net water movement and the cell's final volume relative to its present one. *To check: Cells tab, sucrose, 295 K, preset Plant just goes flaccid.*

3. In the U-tube at 295 K, the left arm holds sucrose and the right arm holds pure water. The two arms together always contain 300 cm of column. At rest the left column stands at 218.6 cm.

(a) Find Ψ_p for each arm. (b) Using the fact that Ψ must match across the membrane, find Ψ_s in the left arm. (c) From that, find the left arm's sucrose concentration in mM. (d) State the right arm's concentration and comment on the pair. *To check: left arm 8.0 mM, right arm 0.0 mM, Refill level, Jump to equilibrium.*

4. An animal cell and a plant cell are both dropped into distilled water. Using $\Psi = \Psi_p + \Psi_s$, explain why the plant cell reaches an equilibrium volume and the animal cell has none available to it. Your answer must say what value Ψ_p is confined to in the animal cell and why that confinement matters.

6 On your own

Nothing here is graded, and none of it appeared above. The demonstration is worth more than this worksheet asked of it.

- Hold the U-tube at 5.0 mM against pure water and drag **Temperature T** from 278 K to 313 K, re-equilibrating at each end. Which way does the equilibrium height difference move, and does the RT value in the hint account for the whole change?
 - Give both arms solute — say 8.0 mM against 3.0 mM. Does the tube settle at the same place as 5.0 mM against 0.0 mM? Decide from the run whether the difference or the ratio of the two concentrations is what governs the endpoint.
 - On the Cells tab with sucrose, find the highest **Bath concentration** at which the animal cell still bursts. Then work out from the lysis threshold quoted in the panel why that particular value is the boundary.
 - Read the panel headed *Where this model stops being true*. Pick the one simplification you think would most change the numbers you recorded today, and say why.
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